Sura Venkata Sailaja

ai23btech11029@iith.ac.in | suravenkatasailaja@gmail.com | +91 9620125712 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.46
Sri Chaitanya Junior College	Class XII (Board of intermediate Eduation,Ap)	2023	98.50%
Sri Chaitanya High School	Class X (Board of secondary education,Andhra pradesh)	2021	98.66%

WORK EXPERIENCE

AI Engineer -- Nikko, Japan:

- Worked as an AI Engineer at Nikko, Japan, developing a prediction model and time-series forecasting system, including a physics-aware model for improved prediction performance.
- Reduced the training time of an existing time-series model from over 8 hours on GPU to around 10 minutes using only CPU, while maintaining similar prediction accuracy, by applying Reservoir Computing and optimizing the modeling approach.
- Implemented and evaluated multiple modeling and optimization methods to build the complete prediction pipeline, integrating physics-aware modeling and data-driven approaches for efficient and accurate forecasting.

MINI PROJECT | Dr.Ashesh

- Implemented SAFMN (ICCV 2023) for ×4 image super-resolution, building the DF2K paired HR/LR pipeline and reproducing the ~239K-parameter architecture.
- Optimized training on an RTX A5000, using batch size 256 with LR = 2×10^-3, and studied data efficiency across 100%, 25%, and 10% training subsets, achieving 30.36 dB PSNR / 0.8355 SSIM
- Developed a train-test distribution-mismatch framework using heavy LR degradations, comparing blind degradation learning with FiLM-based conditioning, where degradation type and severity are injected into feature representations to guide super-resolution.

POSITIONS OF RESPONSIBILITY

SPICMACAY IITH

- Served as Hospitality and Logistics Head, overseeing event responsibilities, artist coordination, and leading the team in managing hospitality and logistics operations.

Sunshine Buddy - Mental Wellness Club:

- Conducted and hosted several student interaction events, facilitating open conversations and engagement among students.

SKILLS

Programming Languages: C, C++, Python(NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), RISC-V

Web Technologies: HTML, CSS

Tools: LaTeX, Git, GitHub, MySQL, SQL, Solid Edge

Soft Skills: Leadership, Good Communication,Teamwork, Time Management

Communication Skills: English, Telugu, Hindi

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Computer Architecture, Database Management Systems, Compilers, Operating Systems

Artificial Intelligence: Computer Vision, Natural Language Processing, Deep Learning, Reinforcement Learning, Foundations of Machine Learning, Probability and Random Variables

Data Science: Data Mining, Predictive Modelling, Applied Mathematics, Bayesian Statistics, Data Science Analysis

PROJECTS

Reliability-Aware Test-Time Adaptation - Dr. Konda Reddy

- Trained a ResNet-18 on clean CIFAR-10 and evaluated test-time adaptation on CIFAR-10-C, using Gaussian Noise, Shot Noise and Brightness corruptions at multiple severity levels and batch sizes 4, 8, 16 and 32.
- Developed a reliability-aware TTA framework combining entropy minimization, entropy-based reliability scoring, selective memory buffering, adaptive loss/learning-rate scaling and reliability-guided Batch Normalization (MEBN). - MEBN consistently reduced classification error; at batch size 32, error fell from 0.3449 to 0.3260 for Gaussian noise, 0.3367 to 0.3208 for Shot noise and 0.3332 to 0.3173 for Brightness compared with TENT

CSI-NER -- Culture-Specific NER for Indian Context - Dr . Maunendra desarkar

- Developed a MuRIL-based NER model for recognizing 7 Indian cultural entity categories.
- Generated and BIO-tagged a synthetic dataset using DIWALI + DOSA with contextual disambiguation.
- Achieved 95.49% F1 on unseen data and 100% F1 on in-domain data.

Grocery Management Web Application - Dr . Shirshendu Das

- Built a grocery-management web app using HTML/CSS, Django and MySQL, supporting customers and sellers.
- Designed a BCNF relational database with customers, sellers, products, orders, ratings and delivery partners.
- Implemented stored procedures, triggers and indexing for order processing, stock validation and efficient queries.

Bayesian Statistics Project - Dr. Arunaba Majumdar

- Applied Bayesian inference on the Students Social Media Addiction dataset using a Beta-Binomial model with a Beta(2,2) prior.
- Obtained posterior Beta(403,201), mean 0.667 and 95% credible interval (0.629, 0.704).
- Also performed Bayesian linear regression on the Student Performance dataset using Metropolis-Hastings, Gibbs sampling, credible intervals and posterior predictive checks.